

Two combinatorial gaps closed: connectivity and odd cycles in the reflection graph $G(\mu)$

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Abstract

Let $G(\mu)$ be the labelled reflection graph attached to a partition $\mu \vdash n$ in [1, §6], whose vertices are the n columns of the local-identity matrix $M^{(\mu)}(t)$ and whose edges are the two-term relations of that paper's Proposition 6.2. Two statements about $G(\mu)$ were left open there, and together they finish that paper's classification theorem. **Both are proved here, and in a sharper form than was asked for.**

- **(G1)** $G(\mu)$ is connected for every μ — indeed it has diameter at most 4, with the corner cell $(1, 1)$ at distance ≤ 2 from every vertex.
- **(G2)** For $n \geq 6$, $G(\mu)$ carries a cycle of nontrivial label product. What is proved is stronger and exponent-free: $G(\mu)$ contains a *triangle* — hence an odd cycle, whose label product is automatically $\neq 1$ — for *every* $\mu \vdash n$ with $n \geq 4$ except exactly $\mu \in \{(2, 2), (3, 2), (2, 2, 1)\}$. Triangle-freeness is classified completely (Theorem 6.4).

The engine is a reformulation (Theorem 3.2): $G(\mu)$ lives on the Young diagram of μ , its edges run only along rows and columns, and

$$(i, k) \sim (i, m) \iff h_{ik} + h_{im} \notin \{\text{hook lengths of row } i\}, \quad (i, k) \sim (i', k) \iff h_{ik} + h_{i'k} \notin \{\text{hook lengths of column } k\}$$

The variable t disappears from the combinatorics entirely. We also supply the one missing proof upstream: [1, Prop. 6.2], the reflection relations themselves, were stated there with a sketch and verified to $n \leq 8$; §2 proves them, by exhibiting the witnessing row of $M^{(\mu)}$ explicitly and showing it has exactly two nonzero entries. Consequently [1, Thm. 3.1] — the classification of the μ for which the Khanna–Loehr local identity with $T = S$ is consistent over $\mathbb{Q}(t)$ — is now **proved for all μ and all n** , not verified to $n \leq 9$, and no step of the chain rests on a computation.

We also **correct** [1, §7(2)]: the obstruction recorded there for $\mu = (3, 1, 1, 1, 1)$ (“no second edge inside row 1 exists”) is false. It arises from using a sufficient condition for a type-(II) edge as if it were necessary. Row 1 of $(3, 1, 1, 1, 1)$ does carry a triangle, exhibited below and confirmed independently from the matrix.

1 Setting and notation

We use the notation of [1] and recall only what is needed. Fix $\mu \vdash n$ with $\ell = \ell(\mu)$ parts and conjugate μ' . Work in the $\mathbb{Z}$ -Maya convention: the bead set is

$$B = B(\mu) = \{b_j := \mu_j - j : j \geq 1\}, \quad \mu_j := 0 \text{ for } j > \ell,$$

so $b_1 > b_2 > \dots$ strictly, and $b_j = -j$ for $j > \ell$. Let $U = \mathbb{Z} \setminus B$ be the set of *holes*. The cell (i, j) of μ has hook length $h_{ij} = \mu_i - j + \mu'_j - i + 1$.

By [1, Prop. 2.1] the columns of $M^{(\mu)}(t)$ — equivalently the elements of $C(\mu) = \{\gamma : \mu/\gamma \text{ a rim hook}\}$ — are in bijection with the pairs (b, b') with $b \in B$, $b' \in U$, $b' < b$, the pair (b, b') corresponding to moving the bead b down to the hole b' ; and $|C(\mu)| = n$. Write $w(b, b')$ for the unknown attached to that column. The edges of $G(\mu)$ are the two-term relations of [1, Prop. 6.2], which we quote verbatim:

Proposition 1.1 ([1, Prop. 6.2]). (I) *Let $c < b$ be beads and $b' < c$ a hole, and suppose $e := b + c - b'$ is a hole (it is automatically $> b$). Then M has a row with exactly the two entries in columns (b, b') and (c, b') , giving $w(c, b') = -t^{|B \cap (c, b)|} w(b, b')$.*

(II) *Let b be a bead and $b' < e < b$ holes, and suppose $c := b' + e - b$ is a bead (it is automatically $< b'$). Then M has a row with exactly the two entries in columns (b, b') and (b, e) , giving $w(b, e) = -t^{1+|B \cap (b', e)|} w(b, b')$.*

Each edge is labelled by its ratio $-t^h \in \mathbb{Q}(t)^\times$. The criterion to be fed is:

Theorem 1.2 ([1, Thm. 6.3]). *If every connected component of $G(\mu)$ contains a cycle whose product of edge labels (traversed with signs) is $\neq 1$ in $\mathbb{Q}(t)$, then $w = 0$ is forced by the rows $\lambda \neq \mu$, and the local identity (1) is inconsistent.*

$$\sum_{L=1}^n \sum_{\gamma \in S(\lambda, L) \cap S(\mu, L)} t^{\text{ht}(\lambda/\gamma)} \tilde{B}(\mu, \gamma) = \delta_{\lambda\mu}, \quad \forall \lambda \vdash n. \quad (1)$$

2 Proposition 1.1 itself, proved

Proposition 1.1 is stated in [1] with a sketch (“sorting the four positions shows that such a λ generically admits exactly two decompositions”) and is verified there for $n \leq 8$; it is not proved. Since it is the only link between $G(\mu)$ and the matrix $M^{(\mu)}$, we prove it here, so that the chain ending in Corollary 6.6 contains no computed step. We use one standard fact, proved in [1, Prop. 2.1]: if $B(\lambda) = B(\gamma) \setminus \{d\} \cup \{d + L\}$ with $d \in B(\gamma)$ and $d + L \notin B(\gamma)$, then λ/γ is a rim hook of size L and $\text{ht}(\lambda/\gamma) = |B(\gamma) \cap (d, d + L)|$; and conversely every rim hook arises so. (The open interval contains neither endpoint, so $B(\gamma)$ may be replaced by $B(\lambda)$.)

Proof of Proposition 1.1. Both cases are the same configuration, and we treat them together. Let $\{b, c\} \subset B$ be two beads and $\{b', e\} \subset U$ two holes with

$$b + c = b' + e.$$

Reflection in the common midpoint exchanges $b \leftrightarrow c$ and $b' \leftrightarrow e$, so after naming $c < b$ and $b' < e$ exactly two orderings are possible: either $b' < c < b < e$ (the beads are nested inside the holes) — this is case (I) — or $c < b' < e < b$ (the holes nested inside the beads) — case (II). Interleavings are impossible: $b' < c$ and $e < b$ would give $b' + e < b + c$, and $c < b'$ with $b < e$ would give $b + c < b' + e$.

Define

$$B_\lambda := (B \setminus \{b, c\}) \cup \{b', e\}, \quad (2)$$

a bead set with the same cofinite left tail as B ; let λ be the corresponding partition. Since $b + c = b' + e$, the sum of the positions is unchanged, so $|\lambda| = |\mu| = n$; and $\lambda \neq \mu$ because $b \in B \setminus B_\lambda$.

Step 1: the two columns. In case (I) put $\gamma_1 = (b, b')$ and $\gamma_2 = (c, b')$; in case (II) put $\gamma_1 = (b, b')$ and $\gamma_2 = (b, e)$. In both cases $\gamma_1, \gamma_2 \in C(\mu)$ by definition (a bead moved down to a hole below it: $b' < b$ always, $b' < c$ in case (I), $e < b$ in case (II)). Writing $B_1 = B \setminus \{b\} \cup \{b'\}$ we get, in either case,

$$B_\lambda = B_1 \setminus \{c\} \cup \{e\},$$

an up-move of the bead c by $e - c = b - b' = |\mu| - |\gamma_1|$. So λ/γ_1 is a rim hook of the right size and $M_{\lambda\gamma_1} \neq 0$. For γ_2 : in case (I), $B_2 = B \setminus \{c\} \cup \{b'\}$ and $B_\lambda = B_2 \setminus \{b\} \cup \{e\}$, an up-move of b by $e - b = c - b' = |\mu| - |\gamma_2|$; in case (II), $B_2 = B \setminus \{b\} \cup \{e\}$ and $B_\lambda = B_2 \setminus \{c\} \cup \{b'\}$, an up-move of c by $b' - c = b - e = |\mu| - |\gamma_2|$ (note $c < b'$ here). Either way $M_{\lambda\gamma_2} \neq 0$.

Step 2: there are no other nonzero entries. Let $\gamma \in C(\mu)$ with $M_{\lambda\gamma} \neq 0$, say $B(\gamma) = B \setminus \{x\} \cup \{y\}$ with $x \in B$, $y \in U$, $y < x$. Then λ/γ is a rim hook, so B_λ and $B(\gamma)$ differ by a single bead move; in particular $|B(\gamma) \setminus B_\lambda| = 1$. Now an element of $B(\gamma)$ lying outside B_λ is either an element of $B \setminus \{x\}$ lying in $\{b, c\}$, or y itself when $y \notin \{b', e\}$ (as $y \notin B$). Hence

$$B(\gamma) \setminus B_\lambda = (\{b, c\} \setminus \{x\}) \cup (\{y\} \text{ if } y \notin \{b', e\}).$$

If $x \notin \{b, c\}$ this set already has two elements, a contradiction. So $x \in \{b, c\}$, and then $\{b, c\} \setminus \{x\}$ is a singleton, forcing $y \in \{b', e\}$. Four pairs (x, y) remain, and the constraint $y < x$ eliminates two of them:

$$\begin{aligned} \text{case (I) } (b' < c < b < e): & \quad (b, b') \checkmark \quad (c, b') \checkmark \quad (b, e), (c, e) \text{ fail since } e > b > c; \\ \text{case (II) } (c < b' < e < b): & \quad (b, b') \checkmark \quad (b, e) \checkmark \quad (c, b'), (c, e) \text{ fail since } c < b' < e. \end{aligned}$$

So $\gamma \in \{\gamma_1, \gamma_2\}$: the row λ of M has exactly two nonzero entries.

Step 3: the exponents. Since $\lambda \neq \mu$, the λ -equation of (1) reads $t^{\text{ht}_1} w(\gamma_1) + t^{\text{ht}_2} w(\gamma_2) = 0$, i.e. $w(\gamma_2) = -t^{\text{ht}_1 - \text{ht}_2} w(\gamma_1)$, where $\text{ht}_j = \text{ht}(\lambda/\gamma_j)$.

Case (I). $\text{ht}_1 = |B_1 \cap (c, e)|$ with $B_1 = B \setminus \{b\} \cup \{b'\}$; the open interval (c, e) contains b but not b' (as $b' < c$), so $\text{ht}_1 = |B \cap (c, e)| - 1$. And $\text{ht}_2 = |B_2 \cap (b, e)|$ with $B_2 = B \setminus \{c\} \cup \{b'\}$; the interval (b, e) contains neither c nor b' , so $\text{ht}_2 = |B \cap (b, e)|$. Hence, using $b \in B$,

$$\text{ht}_1 - \text{ht}_2 = |B \cap (c, e)| - 1 - |B \cap (b, e)| = |B \cap (c, b)] - 1 = |B \cap (c, b)|,$$

giving $w(c, b') = -t^{|B \cap (c, b)|} w(b, b')$.

Case (II). $\text{ht}_1 = |B_1 \cap (c, e)|$; here $c < b' < e < b$, so (c, e) contains b' but not b , whence $\text{ht}_1 = |B \cap (c, e)| + 1$. And $\text{ht}_2 = |B_2 \cap (c, b')|$ with $B_2 = B \setminus \{b\} \cup \{e\}$; the interval (c, b') contains neither b nor e , whence $\text{ht}_2 = |B \cap (c, b')|$. Hence, using that b' is a hole,

$$\text{ht}_1 - \text{ht}_2 = |B \cap (c, e)| + 1 - |B \cap (c, b')| = 1 + |B \cap [b', e)| = 1 + |B \cap (b', e)|,$$

giving $w(b, e) = -t^{1+|B \cap (b', e)|} w(b, b')$. □

Remark 2.1. The proof gives the witnessing row explicitly: (2). It does *not* prove the completeness statement of [1, §6] — that (I) and (II) exhaust the two-term rows of M — and completeness is not needed for Theorem 1.2, which only requires each edge of $G(\mu)$ to be a genuine relation.

3 The structure theorem: $G(\mu)$ is a hook-length graph

Everything below rests on the following dictionary. It is implicit in the proof of [1, Prop. 2.1] but is not stated there, and stating it is what makes the two gaps easy.

Lemma 3.1. *Let $u_1 < u_2 < \dots$ enumerate the holes of $B(\mu)$ in increasing order. Then:*

1. $u_1 = -\ell$; every position $< -\ell$ is a bead, and every position $> b_1$ is a hole.
2. For each $i \leq \ell$, the holes below b_i are exactly $u_1 < \dots < u_{\mu_i}$, and

$$h_{ik} = b_i - u_k \quad (1 \leq k \leq \mu_i).$$

3. Consequently $u_k < b_i \iff k \leq \mu_i \iff i \leq \mu'_k$, and the map $(i, k) \mapsto (b_i, u_k)$ is a bijection from the cells of μ to the vertices of $G(\mu)$.

Proof. (1) The beads are $b_j = \mu_j - j$. For $j > \ell$ we have $b_j = -j \leq -\ell - 1$, so every position $\leq -\ell - 1$ is a bead. The position $-\ell$ is a hole: $b_j = -\ell$ would need $\mu_j = j - \ell$, which is ≤ 0 for $j \leq \ell$ and negative for $j > \ell$. Since b_1 is the largest bead, every position $> b_1$ is a hole.

(2) This is the classical first-column hook-length identity, proved as part of [1, Prop. 2.1]: the set $\{v < b_i : v \notin B\}$ equals $\{b_i - h_{ij} : 1 \leq j \leq \mu_i\}$. As $h_{i1} > h_{i2} > \dots > h_{i\mu_i}$ strictly, the values $b_i - h_{ij}$ increase strictly with j , so listing them in increasing order matches the global enumeration: $b_i - h_{ij} = u_j$. (The count also confirms $u_1 = b_i - h_{i1} = -\ell$ for every i , since $h_{i1} = \mu_i + \ell - i$ and $b_i = \mu_i - i$.)

(3) Immediate from (2): there are exactly μ_i holes below b_i , namely $u_1, \dots, u_{\mu_i}$; and $k \leq \mu_i \iff i \leq \mu'_k$ is the definition of the conjugate. Summing over i gives n pairs, matching $|C(\mu)| = n$. $\square$

We henceforth name vertices by cells. Write

$$H_i = \{h_{i1}, \dots, h_{i\mu_i}\} \quad (\text{hooks of row } i), \quad H^{(k)} = \{h_{1k}, \dots, h_{\mu'_k, k}\} \quad (\text{hooks of column } k).$$

Theorem 3.2 (structure). *The vertex set of $G(\mu)$ is the set of cells of μ , and every edge joins two cells in the same row or two cells in the same column. Precisely, for $k \neq m$ and $i \neq i'$:*

$$(i, k) \sim (i, m) \text{ (type II)} \iff h_{ik} + h_{im} \notin H_i, \quad (3)$$

$$(i, k) \sim (i', k) \text{ (type I)} \iff h_{ik} + h_{i'k} \notin H^{(k)}. \quad (4)$$

There are no other edges. The label of the edge (3) is $-t^{1+|B \cap (u_k, u_m)|}$ and of (4) is $-t^{|i' - i| - 1}$.

Proof. A type-(II) relation fixes the bead b and varies the hole; by Lemma 3.1 that is two cells in the same row. A type-(I) relation fixes the hole b' and varies the bead: two cells in the same column. As (I) and (II) are the only relations in Proposition 1.1, there are no other edges.

Type (II). Take $b = b_i$ and holes $u_k < u_m < b_i$, i.e. $1 \leq k < m \leq \mu_i$. Using $u_k = b_i - h_{ik}$ and $u_m = b_i - h_{im}$,

$$c = u_k + u_m - b_i = b_i - (h_{ik} + h_{im}).$$

Since $h_{im} \geq 1$ we have $c < u_k < b_i$, so c is a hole if and only if c is one of the holes below b_i , i.e. $c = b_i - h_{iq}$ for some $q \leq \mu_i$, i.e. $h_{ik} + h_{im} \in H_i$. (If $h_{ik} + h_{im} > h_{i1}$ then $c < -\ell$ and c is a bead by Lemma 3.1(1); this is the case $h_{ik} + h_{im} \notin H_i$ as well.) The edge exists exactly when $c \in B$, giving (3).

Type (I). Take beads $b = b_i > c = b_{i'}$, so $i < i'$, and a hole $u_k < b_{i'}$, i.e. $k \leq \mu_{i'}$, equivalently $i' \leq \mu'_k$. Then

$$e = b_i + b_{i'} - u_k = u_k + h_{ik} + h_{i'k},$$

and $e > b_i$ since $h_{i'k} \geq 1$. Every bead is some b_j ; and $e > u_k \geq -\ell$ forces $j \leq \ell$, while $e > u_k$ forces $j \leq \mu'_k$ by Lemma 3.1(3). So e is a bead if and only if $u_k + h_{ik} + h_{i'k} = u_k + h_{jk}$ for some $j \leq \mu'_k$, i.e. $h_{ik} + h_{i'k} \in H^{(k)}$. The edge exists exactly when $e \in U$, giving (4). For its label, $|B \cap (b_{i'}, b_i)| = |\{b_{i+1}, \dots, b_{i'-1}\}| = i' - i - 1$. $\square$

Remark 3.3. Theorem 3.2 is manifestly self-conjugate: conjugation of μ acts on the abacus by $x \mapsto -1 - x$, exchanging beads with holes, hence exchanging type (I) with type (II), and on cells by transposition, which exchanges (3) with (4). Every statement below therefore comes in a row/column pair, and we prove only one of each.

4 (G1): connectivity

Lemma 4.1 (hubs). *For every i and every $2 \leq m \leq \mu_i$ the edge $(i, 1) \sim (i, m)$ is present; for every k and every $2 \leq i' \leq \mu'_k$ the edge $(1, k) \sim (i', k)$ is present.*

Proof. Hook lengths decrease strictly along a row, so $h_{i1} = \max H_i$. Hence $h_{i1} + h_{im} > h_{i1} = \max H_i$, so $h_{i1} + h_{im} \notin H_i$, and (3) gives the edge. Hook lengths decrease strictly down a column, so $h_{1k} = \max H^{(k)}$ and the same one-line argument with (4) gives the second statement. $\square$

Theorem 4.2 ((G1)). *$G(\mu)$ is connected for every partition μ . More precisely every vertex is within distance 2 of the corner cell $(1, 1)$, so $\text{diam } G(\mu) \leq 4$.*

Proof. Let (i, k) be a cell. By Lemma 4.1 applied to row i , $(i, k) \sim (i, 1)$ (or $k = 1$). By Lemma 4.1 applied to column 1, $(i, 1) \sim (1, 1)$ (or $i = 1$). So $d((i, k), (1, 1)) \leq 2$. $\square$

This is exactly the “hub” idea sketched in [1, §7(2)], but with the *exact* edge condition (3) in place of the sufficient condition used there, and applied to every row rather than only row 1.

5 Odd cycles have nontrivial label product

Lemma 5.1. *Every edge label of $G(\mu)$ has the form $-t^h$ with $h \in \mathbb{Z}$. If a cycle has odd length ℓ , its label product is $-t^m$ for some $m \in \mathbb{Z}$, hence $\neq 1$ in $\mathbb{Q}(t)$.*

Proof. In $\mathbb{Q}(t)^\times$ we have $(-t^h)(-t^{-h}) = t^0 = 1$, so $(-t^h)^{-1} = -t^{-h}$: traversing an edge backwards inverts the exponent but *does not* change the sign. Hence each of the ℓ edges of the cycle contributes exactly one factor -1 , and the product is $(-1)^\ell t^{\sum_j \pm h_j}$. For ℓ odd this is $-t^m$. If $-t^m = 1$ in $\mathbb{Q}(t)$, evaluate at $t = 1$ to get $-1 = 1$, a contradiction. $\square$

Corollary 5.2. *If $G(\mu)$ is connected and not bipartite, the hypothesis of Theorem 1.2 holds and (1) is inconsistent over $\mathbb{Q}(t)$. No exponent bookkeeping is required.*

Proof. A connected non-bipartite graph contains an odd cycle; apply Lemma 5.1 and Theorem 4.2. $\square$

6 (G2): classification of triangle-free $G(\mu)$

Lemma 6.1 (line lemma). *Let $h_1 > h_2 > \dots > h_r \geq 1$ be integers and put $S = \{h_1, \dots, h_r\}$. Call a pair $2 \leq q < s \leq r$ good if $h_q + h_s \notin S$. Then:*

1. *If $\{q, s\}$ is good, the three indices $\{1, q, s\}$ are pairwise joined in the sense of (3) (resp. (4)).*
2. *A good pair exists if and only if $r \geq 4$, or $r = 3$ and $h_2 + h_3 \neq h_1$.*

Proof. (1) The pairs $\{1, q\}$ and $\{1, s\}$ are good by the argument of Lemma 4.1 ($h_1 + h_x > h_1 = \max S$), and $\{q, s\}$ is good by hypothesis.

(2) First, if $h_q + h_s \in S$, say $h_q + h_s = h_j$, then $h_j > h_q$ forces $j < q$; in particular for $q = 2$ we get $j = 1$, i.e. $h_2 + h_s = h_1$.

If $r \leq 2$ there is no pair at all. If $r = 3$ the only pair is $\{2, 3\}$, good iff $h_2 + h_3 \neq h_1$. If $r \geq 4$, consider the two pairs $\{2, 3\}$ and $\{2, 4\}$. Were both bad we would have $h_2 + h_3 = h_1 = h_2 + h_4$, hence $h_3 = h_4$, contradicting strictness. So one of them is good. $\square$

Lemma 6.2. *Every triangle of $G(\mu)$ lies inside a single row or a single column of μ .*

Proof. By Theorem 3.2 each edge lies in a row or a column. Suppose a triangle $\{x, y, z\}$ has x, y in row i and x, z in column k , with $x = (i, k)$, $y = (i, k_y)$, $z = (i_z, k)$, $k_y \neq k$, $i_z \neq i$. Then y and z lie in different rows and different columns, so no edge joins them. Hence all three edges are of the same kind and share a row, or share a column. $\square$

Lemma 6.3. *Row 1 of μ contains no triangle if and only if $\mu_1 \leq 2$, or $\mu_1 = 3$ and $n = 2\ell + 1$. Dually, column 1 contains no triangle if and only if $\ell \leq 2$, or $\ell = 3$ and $n = 2\mu_1 + 1$.*

Proof. Apply Lemma 6.1 with $r = \mu_1$ and $h_j = h_{1j}$. A triangle inside row 1 exists iff a good pair exists. By Lemma 6.1(2) this fails iff $\mu_1 \leq 2$, or $\mu_1 = 3$ and $h_{12} + h_{13} = h_{11}$.

Suppose $\mu_1 = 3$. Then $h_{11} = \mu_1 + \mu'_1 - 1 = \ell + 2$, $h_{12} = \mu_1 - 2 + \mu'_2 - 1 + 1 = \mu'_2 + 1$, and $h_{13} = \mu_1 - 3 + \mu'_3 - 1 + 1 = \mu'_3$. Also $n = \mu'_1 + \mu'_2 + \mu'_3 = \ell + \mu'_2 + \mu'_3$ because $\mu_1 = 3$. Hence

$$h_{12} + h_{13} = h_{11} \iff \mu'_2 + \mu'_3 + 1 = \ell + 2 \iff (n - \ell) = \ell + 1 \iff n = 2\ell + 1.$$

The column statement is the conjugate. $\square$

Theorem 6.4 (classification of triangle-free $G(\mu)$). *$G(\mu)$ contains no triangle if and only if $n \leq 3$ or $\mu \in \{(2, 2), (3, 2), (2, 2, 1)\}$.*

Proof. ($\Leftarrow$) For $n \leq 3$ and for the three listed μ every row and every column of μ has at most 3 cells, and where it has exactly 3 one checks $h_2 + h_3 = h_1$ directly; by Lemmas 6.1 and 6.2 there is no triangle. Explicitly: for $\mu = (3)$ the row-1 hooks are 3, 2, 1 with $2 + 1 = 3$; for $\mu = (2, 2)$ and $\mu = (2, 1)$ no line has 3 cells; for $\mu = (3, 2)$ the row-1 hooks are 4, 3, 1 with $3 + 1 = 4$ and every other line has ≤ 2 cells; (1^3) , $(1, 1)$, $(2, 2, 1)$ are the conjugates of (3) , (2) , $(3, 2)$.

($\Rightarrow$) Assume $n \geq 4$ and $G(\mu)$ triangle-free. Then by Lemma 6.3 both

$$(\mu_1 \leq 2 \text{ or } (\mu_1 = 3 \text{ and } n = 2\ell + 1)) \quad \text{and} \quad (\ell \leq 2 \text{ or } (\ell = 3 \text{ and } n = 2\mu_1 + 1)).$$

In particular $\mu_1 \leq 3$ and $\ell \leq 3$, so $n \leq 9$. Four cases.

(a) $\mu_1 \leq 2$ and $\ell \leq 2$. Then $n \leq 4$, so $n = 4$ and $\mu = (2, 2)$.

(b) $\mu_1 \leq 2$, $\ell = 3$, $n = 2\mu_1 + 1$. If $\mu_1 = 1$ then $n = 3$, excluded. So $\mu_1 = 2$, $n = 5$, $\ell = 3$: $\mu = (2, 2, 1)$.

(c) $\mu_1 = 3$, $n = 2\ell + 1$, $\ell \leq 2$. If $\ell = 1$ then $n = 3$, excluded. So $\ell = 2$, $n = 5$, $\mu_1 = 3$: $\mu = (3, 2)$.

(d) $\mu_1 = 3$, $\ell = 3$, $n = 2\ell + 1 = 7$. Then $\mu \vdash 7$ with $\mu_1 = 3$, $\ell = 3$, so $\mu = (3, 3, 1)$ or $\mu = (3, 2, 2)$. We rule both out using a *second* line. For $\mu = (3, 3, 1)$ we have $\mu' = (3, 2, 2)$ and the row-2 hooks are

$$h_{21} = 3 - 1 + 3 - 2 + 1 = 4, \quad h_{22} = 3 - 2 + 2 - 2 + 1 = 2, \quad h_{23} = 3 - 3 + 2 - 2 + 1 = 1,$$

so $h_{22} + h_{23} = 3 \notin \{4, 2, 1\} = H_2$ and $\{(2, 1), (2, 2), (2, 3)\}$ is a triangle by Lemma 6.1(1). The case $\mu = (3, 2, 2) = (3, 3, 1)'$ follows by conjugation (its column-2 hooks are likewise 4, 2, 1).

So the only triangle-free μ with $n \geq 4$ are $(2, 2), (3, 2), (2, 2, 1)$. $\square$

Corollary 6.5 ((G2), in strengthened form). *For every $\mu \vdash n$ with $n \geq 4$ and $\mu \notin \{(2, 2), (3, 2), (2, 2, 1)\}$, the graph $G(\mu)$ contains a triangle. Hence it contains an odd cycle, whose label product is $\neq 1$ by Lemma 5.1. In particular this holds for every μ with $n \geq 6$, which is (G2).*

Corollary 6.6. *[1, Thm. 3.1] is proved, for all μ and all n : the system (1) is consistent over $\mathbb{Q}(t)$ exactly for $n \leq 3$ (all μ), for $n = 4$ with $\mu = (2, 2)$, and for $n = 5$ with $\mu \in \{(3, 2), (2, 2, 1)\}$; and inconsistent otherwise.*

Proof. Inconsistency for $n \geq 4$, $\mu \notin \{(2, 2), (3, 2), (2, 2, 1)\}$: combine Theorem 4.2, Corollary 6.5 and Theorem 1.2 (whose hypothesis Proposition 1.1 is proved in §2).

Consistency in the nine remaining cases is proved by exhibiting a solution; each is checked by substitution into (1), and each denominator below is a nonzero element of $\mathbb{Q}(t)$. Listing w against the columns $\gamma \in C(\mu)$ in the order shown:

$$\begin{aligned}
\mu = (1) : & \quad w_\emptyset = 1 \\
\mu = (2) : & \quad \frac{1}{t-1}(t, -1) \text{ on } ((1), \emptyset) \\
\mu = (1^2) : & \quad \frac{1}{t-1}(-1, 1) \text{ on } ((1), \emptyset) \\
\mu = (3) : & \quad \frac{1}{2t-1}(t, t, -1) \text{ on } ((2), (1), \emptyset) \\
\mu = (2, 1) : & \quad \frac{1}{t^2-t+1}(1, t^2, -1) \text{ on } ((2), (1, 1), \emptyset) \\
\mu = (1^3) : & \quad \frac{1}{t(t-2)}(-t, -1, 1) \text{ on } ((1, 1), (1), \emptyset) \\
\mu = (2, 2) : & \quad \frac{1}{(t-1)^2}(-t, 1, t, -1) \text{ on } ((2, 1), (2), (1, 1), (1)) \\
\mu = (3, 2) : & \quad \frac{1}{2t^2-2t+1}(-t, t^2, 1, t, -1) \text{ on } ((3, 1), (2, 2), (3), (1, 1), (1)) \\
\mu = (2, 2, 1) : & \quad \frac{1}{t(t^2-2t+2)}(t, -t^2, t^2, 1, -1) \text{ on } ((2, 2), (2, 1, 1), (1^3), (2), (1))
\end{aligned}$$

Remark 6.7. Corollary 6.6 does not rely on the $n \leq 9$ computation of [1, §6]; that computation $\square$ now serves purely as a check. The hypothesis of [1, Prop. 5.2], $\text{rank } M^{(\mu)}(-1) = n$, is *not* used anywhere above and remains open.

7 Correction to [1, §7(2)]

That paper records, as an obstruction to finding triangles in row 1:

“a type-(II) edge between two cells of row i with hooks $d_1 > d_2$ exists as soon as $d_1 + d_2 > h_{i1}$ — so the corner cell $(1, 1)$ is joined to every other cell of row 1. That gives a hub but not, by itself, a triangle: for $\mu = (3, 1, 1, 1, 1)$ the row-1 hooks are 7, 2, 1 and no second edge inside row 1 exists.”

The condition $d_1 + d_2 > h_{i1}$ is *sufficient* but not necessary. By (3) the exact condition is $d_1 + d_2 \notin H_i$, and $d_1 + d_2 > h_{i1} = \max H_i$ is only one way for that to happen.

Example 7.1 ($\mu = (3, 1, 1, 1, 1)$). Here $n = 7$, $\ell = 5$, $\mu' = (5, 1, 1)$; the row-1 hooks are $H_1 = \{7, 2, 1\}$. Then

$$h_{12} + h_{13} = 2 + 1 = 3 \notin \{7, 2, 1\} = H_1,$$

so by (3) the edge $(1, 2) \sim (1, 3)$ **exists**, and with the two hub edges of Lemma 4.1 the set $\{(1, 1), (1, 2), (1, 3)\}$ is a triangle. Its labels are $-t^5, -t^5, -t$: on the abacus $b_1 = 2$, the holes below it are $-5 < 0 < 1$, and the three relations read

$$w(2, 0) = -t^5 w(2, -5), \quad w(2, 1) = -t^5 w(2, -5), \quad w(2, 1) = -t w(2, 0).$$

Substituting the first into the third and comparing with the second gives $-t^5(1+t)w(2, -5) = 0$, so $w(2, -5) = 0$; with Theorem 4.2 the whole solution vanishes. Consistently with Lemma 5.1, the label product around the triangle is $(-t^5)(-t)(-t^5)^{-1} = -t^6 \neq 1$.

This is not a repair of a false theorem: [1, Thm. 6.3] and [1, Prop. 6.2] are correct, and the sentence quoted is a remark in the gaps section. But the remark was the stated reason for not pursuing triangles, and it is void. Note also that the sufficient condition suffices for the hub statement, so Lemma 4.1 — and hence (G1) — was already within reach in [1]; what was missing was the exact condition, which is what (G2) needs.

8 Verification

All scripts are in `proofs/scripts-2026-09-11-Q143/`.

Engines. Three independent constructions of $G(\mu)$ were built and compared.

- **E1** (*deriver*, `graph.py`): from Proposition 1.1 on the $\mathbb{Z}$ -Maya abacus. Asserts $|V(G(\mu))| = n$ for every μ tested.
- **E1b** (*deriver*, `hookgraph.py`): from Theorem 3.2, using hook lengths only — no abacus.
- **E2** (*comparator* / *answer key*, `matrix_engine.py`): builds $M^{(\mu)}(t)$ from rim-hook combinatorics on Young diagrams (connectivity and no- 2×2 tests on cell sets), then extracts the rows with exactly two nonzero entries. This uses neither the abacus nor hook lengths.

Results: **E1** = **E1b** on all 138 partitions with $n \leq 10$ (edge sets identical). **E1** = **E2** on all 66 partitions with $n \leq 8$, agreeing on edge sets *and* on labels. This is sharper than the claim recorded in [1], which asserted only that the two families span the same space.

Main table. Per n , over all $\mu \vdash n$: number of μ with $G(\mu)$ connected; with $G(\mu)$ non-bipartite; for which the criterion of Theorem 1.2 holds (tested on a spanning forest's fundamental cycles, which generate the cycle space).

n	$p(n)$	connected	non-bipartite	criterion	triangle-free μ
1	1	1	0	0	(1)
2	2	2	0	0	(2), (1 ²)
3	3	3	0	0	(3), (2, 1), (1 ³)
4	5	5	4	4	(2, 2)
5	7	7	5	5	(3, 2), (2, 2, 1)
6	11	11	11	11	—
7	15	15	15	15	—
8	22	22	22	22	—
9	30	30	30	30	—
Σ	96	96	87	87	9

The totals factorise as $96 = 1 + 2 + 3 + 5 + 7 + 11 + 15 + 22 + 30$ and $87 = 96 - 9$, the 9 being the six partitions with $n \leq 3$ together with $(2, 2), (3, 2), (2, 2, 1)$ — exactly the exceptional set of Theorem 6.4, and exactly the set on which (1) is consistent. ([1] reports “95 partitions, $n \leq 9$ ”; that count omits $n = 1$.) The columns “non-bipartite” and “criterion” agree entry by entry at every n : no even cycle with nonzero signed exponent sum is ever needed.

The boundary population, explicitly. A negative statement needs its counterexamples exhibited. The nine triangle-free cases: $(1), (2), (1^2), (3), (2, 1), (1^3)$ give trees ($|E| = |V| - 1$, so no cycle at all), while

$$(2, 2) : |V| = 4, |E| = 4; \quad (3, 2) : |V| = 5, |E| = 5; \quad (2, 2, 1) : |V| = 5, |E| = 5,$$

each having exactly one independent cycle, of *even* length 4 and signed exponent sum 0, hence label product $(-1)^4 t^0 = 1$. For $(2, 2)$ the cycle is $(1, 1) \rightarrow (2, 1) \rightarrow (2, 2) \rightarrow (1, 2) \rightarrow (1, 1)$ with exponents 0, +1, -0, -1.

Checks of the proved statements.

- Lemma 4.1: all hub edges $(i, 1) \sim (i, m)$ and $(1, k) \sim (i', k)$ present, for every μ with $n \leq 10$. 0 failures.
- Lemma 6.1: the predicate “ $r \geq 4$, or $r = 3$ and $h_2 + h_3 \neq h_1$ ” agrees with direct search for a good pair on all 1624 lines (rows and columns) of all partitions with $n \leq 11$. 0 failures.
- Lemma 6.3: “row 1 triangle-free $\iff \mu_1 \leq 2$ or $(\mu_1 = 3 \text{ and } n = 2\ell + 1)$ ” holds for all 271 partitions with $n \leq 12$. 0 failures.
- Theorem 6.4: the classification agrees with a brute-force search over all 3-subsets of cells (not using Lemma 6.1) on all 271 partitions with $n \leq 12$. 0 failures. Over $n \leq 10$, triangle-free $\iff$ bipartite.
- Proposition 1.1 (§2): for every μ with $n \leq 8$ and every configuration $b+c = b'+e$ of two beads and two holes, the row λ of (2) was built and looked up in the matrix of engine E2. All 596 configurations give $\lambda \neq \mu$ with $|\lambda| = n$, exactly two nonzero entries, in the predicted columns, with the predicted exponent difference. 0 failures. The count factorises as $596 = 298 + 298$ (type (I) / type (II), equal at every n by conjugation) $= 0 + 2 + 6 + 20 + 38 + 90 + 150 + 290$ over $n = 1, \dots, 8$, and equals the summed edge count $\sum_{\mu} |E(G(\mu))|$ — each configuration contributes exactly one edge, with no coincidences.
- Corollary 6.6: the nine solution vectors were obtained by solving $Mw = e_{\mu}$ over $\mathbb{Q}(t)$ with SYMPY and then *substituted back*; all nine satisfy $Mw - e_{\mu} = 0$ identically. In the other direction, for every non-exceptional μ with $4 \leq n \leq 6$, $\text{rank}[M \mid e_{\mu}] = \text{rank } M + 1$, confirming inconsistency independently of the graph criterion.
- Example 7.1: the edge $(1, 2) \sim (1, 3)$ of $G((3, 1, 1, 1, 1))$ is exhibited independently by E2 — the row $\lambda = (2, 2, 1, 1, 1)$ of $M^{((3, 1, 1, 1, 1))}$ has exactly two nonzero entries, in columns $\gamma = (2, 1, 1, 1, 1)$ (entry t^0) and $\gamma = (1, 1, 1, 1, 1)$ (entry t^1), giving $w_{(1^5)} = -t^{-1}w_{(2, 1^4)}$.

Negative controls. Each perturbation must move a reported number; one that does not would invalidate the control rather than confirm the claim.

1. *Delete all type-(I) edges.* Connectivity collapses: over $n \leq 10$ the count of connected $G(\mu)$ falls from 96 to 9 (only $\mu = (n)$ survives at each n). **Moved.**
2. *Delete all type-(II) edges.* Same totals, but by conjugation, not by being the same test: here only $\mu = (1^n)$ survives. Checked per μ : control 1 on μ and control 2 on μ' agree in ($\#$ components, non-bipartite) for all μ , $n \leq 9$; and they differ on a single μ , e.g. $\mu = (3, 1)$ gives 2 components versus 3. **Moved.**
3. *Replace the exact type-(II) condition (3) by the sufficient condition $d_1 + d_2 > h_{i1}$ of [1, §7(2)].* The non-bipartite count moves at $n = 4$ ($4 \rightarrow 3$), $n = 5$ ($5 \rightarrow 4$) and $n = 7$ ($15 \rightarrow 14$). **Moved** — this is the control that isolates the error corrected in §7.
4. *Drop the $r = 3$ exception from Lemma 6.1, i.e. predict “a good pair exists iff $r \geq 3$ ”.* 56 wrong predictions over the lines with $n \leq 11$, the smallest being row 1 of (3) with hooks 3, 2, 1; the correct form makes 0. **Moved.**

9 What is *not* proved

1. The rank hypothesis $\text{rank } M^{(\mu)}(-1) = n$ of [1, Prop. 5.2] — linear independence of $\{p_{n-|\gamma|} s_\gamma : \gamma \in C(\mu)\}$ in Λ_n — is untouched here. It is verified for $n \leq 8$ and is not used above.
2. Gap (2) of the earlier paper: Khanna–Loehr allow $T(\mu, L)$ to be an arbitrary subset of $R(n - L)$. Nothing here bears on that; Corollary 6.6 refutes only the symmetric choice $T = S$.
3. **Completeness** of Proposition 1.1: the claim in [1, §6] that (I) and (II) span the same space as *all* two-term rows of M is *not* proved here — §2 proves only that each (I)/(II) configuration is a genuine two-term row, which is all Theorem 1.2 needs. Completeness is verified in the stronger form “the two sets of relations coincide, labels included” for $n \leq 8$ (§8). It would be needed only for a converse to Theorem 1.2.
4. Converse direction of Theorem 1.2: failing the cycle criterion is not shown to imply consistency. The agreement of the last two columns of the table is a computed coincidence over $n \leq 9$, not a theorem. Corollary 6.6 does not need it, since consistency in the nine exceptional cases is established by exhibiting solutions.

References

- [1] Clio, *A certificate family for the Khanna–Loehr local identity, and a graph criterion for general μ* , proofs/2026-09-10-c2-Q140-local-identity-certificate-family.tex, 10 September 2026.